

Statistics

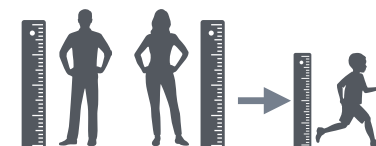
Characterizing observed randomness

Sir Francis Galton
(1822-1911)



Case of study:
Human height heritage

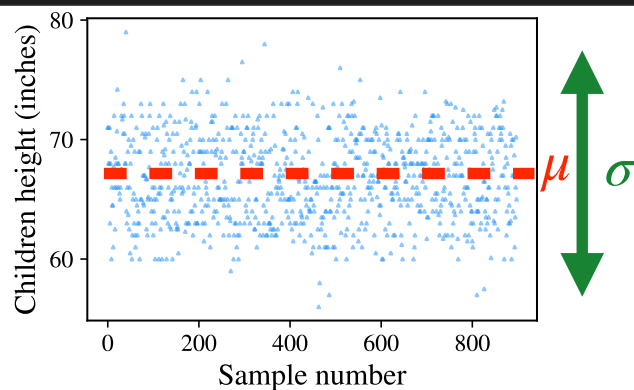
Mean, variance, histogram, correlations



Data

	family	father	mother	gender	height	kids	male	female
0	1	78.5	67.0	M	73.2	4	1.0	0.0
1	1	78.5	67.0	F	69.2	4	0.0	1.0
2	1	78.5	67.0	F	69.0	4	0.0	1.0
3	1	78.5	67.0	F	69.0	4	0.0	1.0
4	2	75.5	66.5	M	73.5	4	1.0	0.0
...	...	...	...	...	...	...	...	...
893	136A	68.5	65.0	M	68.5	8	1.0	0.0
894	136A	68.5	65.0	M	67.7	8	1.0	0.0
895	136A	68.5	65.0	F	64.0	8	0.0	1.0
896	136A	68.5	65.0	F	63.5	8	0.0	1.0
897	136A	68.5	65.0	F	63.0	8	0.0	1.0

898 rows x 8 columns



Summary statistics

measures: N

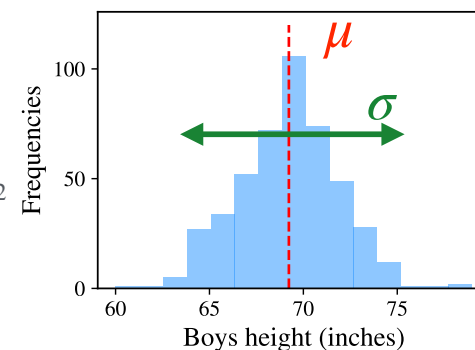
Sample Mean: $\hat{\mu} = \frac{1}{N} \sum_{i=1}^N x_i$

measures:
 $x_1, x_2, \dots, x_N$

Sample Variance: $\hat{\sigma}^2 = \frac{1}{N} \sum_{i=1}^N (x_i^2 - \hat{\mu})^2$

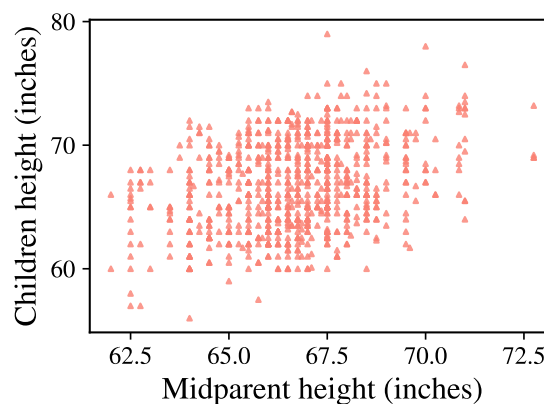
Frequency:
(histogram)

$f(x) = \sum_{i=1}^N I(x_i \in [x, x + dx])$



Correlations

$\hat{\rho} \sim 0.3$



“Midparent” height : $h_m = \frac{1}{2} (h_{father} + h_{mother})$

Sample Covariance: $\hat{C} = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{\mu}_x)(y_i - \hat{\mu}_y)$

Pearson's coefficient: $\hat{\rho} = \frac{\hat{C}}{\hat{\sigma}_x \hat{\sigma}_y} \in [-1, 1]$

$\hat{\rho} > 0$ The two variables tend to increase together

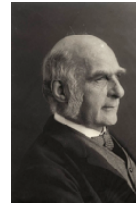
$\hat{\rho} < 0$ If one increases the other tend to decrease

$\hat{\rho} \approx 0$ Decorrelated variables

Statistics

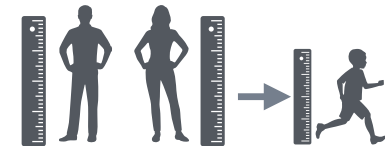
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Linear regression

Best straight line fit to data

$$y = \alpha x + \beta$$

$$\alpha = \frac{\hat{C}}{\hat{\sigma}_x^2} = \rho \frac{\hat{\sigma}_y}{\hat{\sigma}_x}$$

$$\beta = \hat{\mu}_y - \alpha \hat{\mu}_x$$

$$\alpha = \frac{2}{3}$$

Variability controlled
through generations

Regression to mediocrity

